

Product Brochure on IoT

The Power of IoT Benefits for Modern Businesses

Developed by Professors of IISc Bangalore, IIT Patna and NIT Rourkela and selling by Ewarn system

Empowering enterprises with cutting-edge IoT technologies, sensor systems, and data analytics solutions that drive digital transformation and create sustainable business value.





EWARN Hands-on Laboratory Ecosystem

Developed by Professors of IISc Bangalore, IIT Patna and NIT Rourkela and selling by Ewarn system

Empowering Innovation | Building Future Skills | Driving Research Excellence

This brochure offers an in-depth look at the EWARN Hands-on Laboratory Ecosystem, designed to empower innovation, skill development, and research excellence. It presents detailed module descriptions, technical specifications, and competitive bulk order pricing, along with an exclusive partner pricing section for our authorized marketing and distribution associates.

Each kit is a complete, ready-to-use solution, fully supported with comprehensive documentation, step-by-step manuals, verified source codes, and guided experiments ensuring a seamless learning and implementation experience. With EWARN, institutions and partners can bring cutting-edge technology directly into their classrooms and labs, accelerating both education and innovation.

Step into the future of learning with the EWARN Hands-on Laboratory Ecosystem a complete, ready-to-use solution designed for colleges, universities, and research institutions. Developed with a focus on emerging technologies like IoT, 5G/6G, AI/ML, and smart sensing, our ecosystem brings industry-grade tools and knowledge directly into the classroom.

What's Inside:

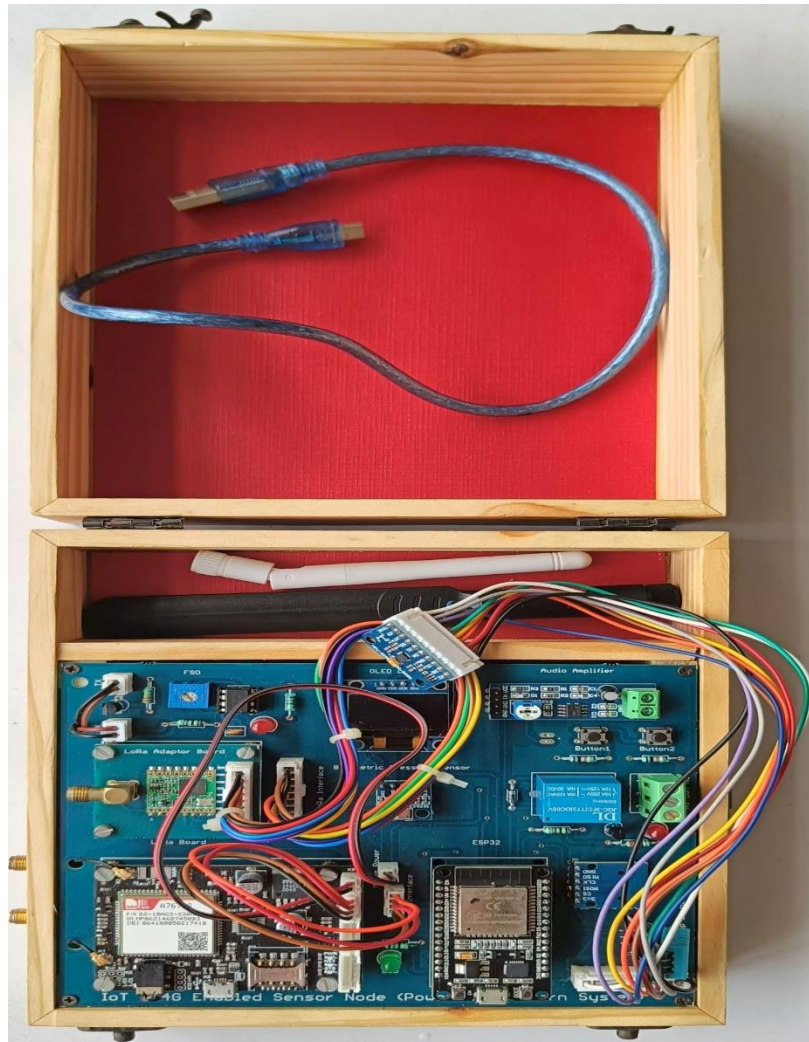
- ✓ Cutting-edge Modules: Detailed descriptions, specifications, and real-world use cases
- ✓ Flexible Pricing: Competitive bulk order rates & exclusive partner pricing for authorized distributors
- ✓ All-in-One Learning Kit: Documentation, step-by-step manuals, verified source codes, and guided experiments

Why Choose EWARN:

- ✓ Aligns with AICTE-recommended curricula
 - ✓ Supports multiple communication protocols: Wi-Fi, Bluetooth, LoRa, GSM, GPRS, and FSO
 - ✓ Accelerates full-stack product development in both hardware and software
- With EWARN, you don't just teach technology, you create innovators.

Partner with us today and bring cutting-edge tech education to your institution or distribution network.

Product 1: All-in-One IoT Development Kit (ESP32) with inbuilt sensors and actuators



Description: A powerful ESP32-based development platform that enables students to learn IoT, cloud computing, and communication protocols. It supports 30+ experiments aligning with AICTE IoT lab curriculum. Includes interfaces such as Wi-Fi, GSM, LoRA, Bluetooth, and FSO.

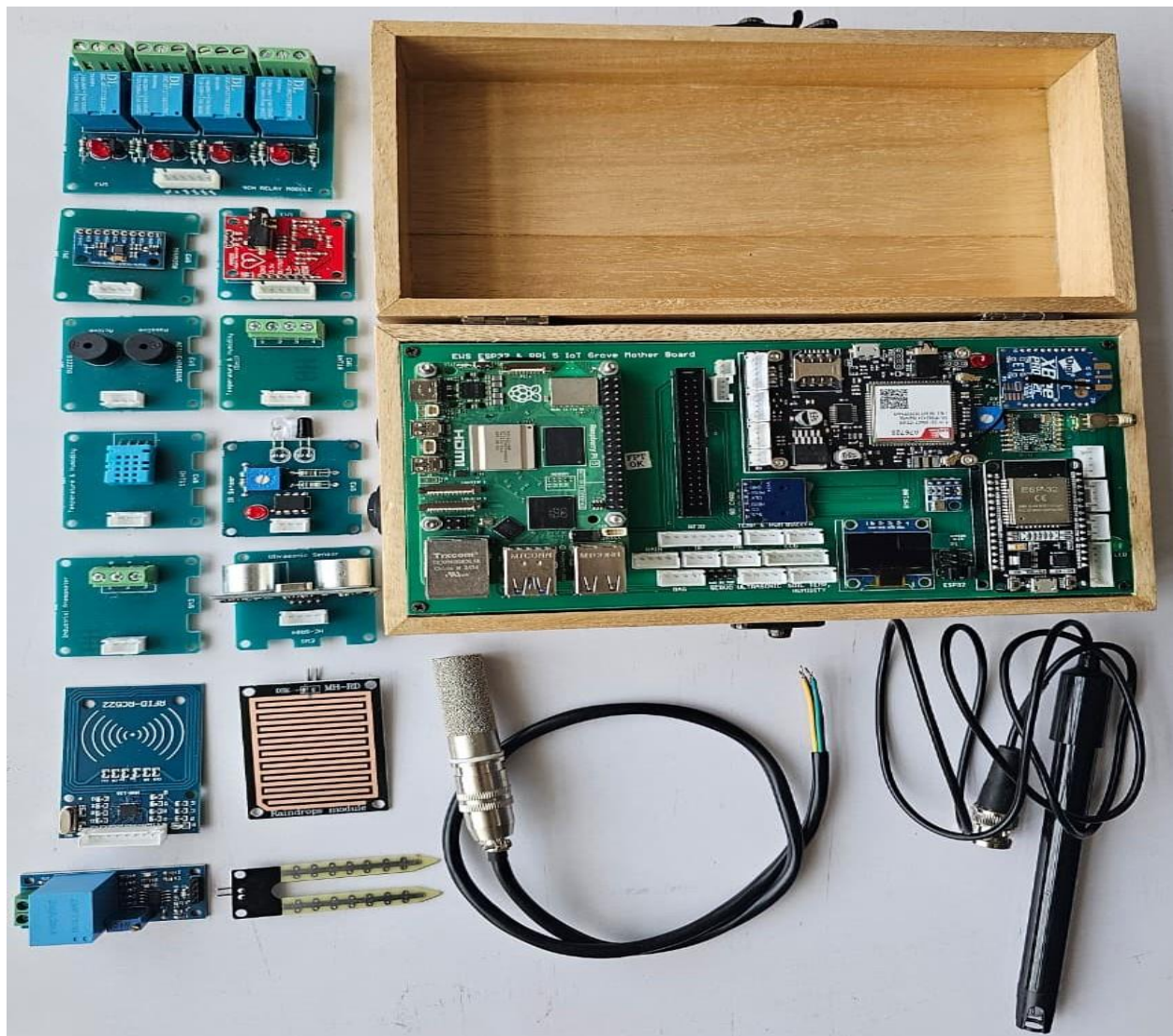
Specifications:

Description: A plug-and-play solution built on Arduino Mega and Grove modules, featuring more than 60+ experiments tailored for agriculture, smart home, robotics, and environmental monitoring.

Specifications: **Micro Controller:** ESP32, **Network Interfaces:** Wi-Fi, Bluetooth, GSM/ GPS/GPRS, LoRA, FSO, **Sensors:** DHT11, BMP180, IMU, Rain Sensor, Ultrasonic, **Actuators:** Relay, Speaker, Buzzer, **Display:** OLED, **Storage:** SD Card.

Number of Experiments: 60+ Documents Included: Experiment Manual, Codes, Setup Instructions.

Product 3: Raspberry Pi + ESP32 Grove IoT Kit

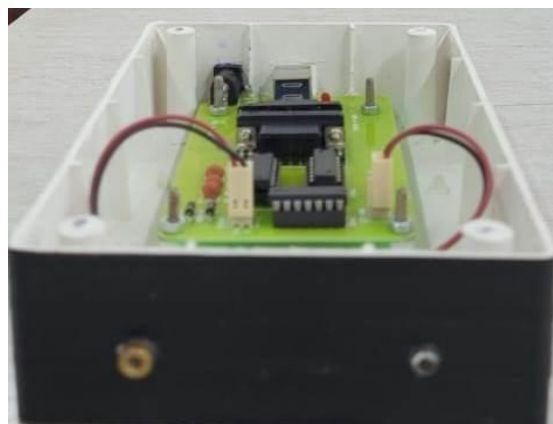
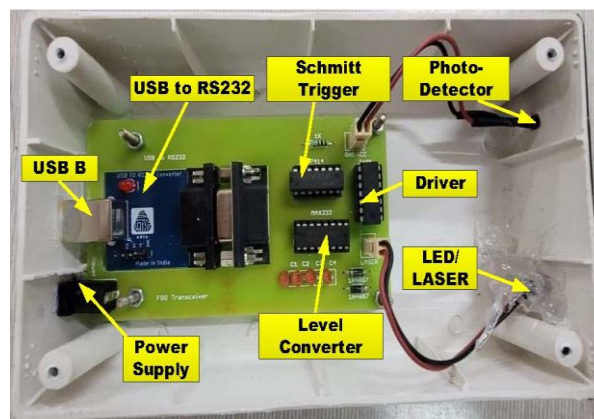


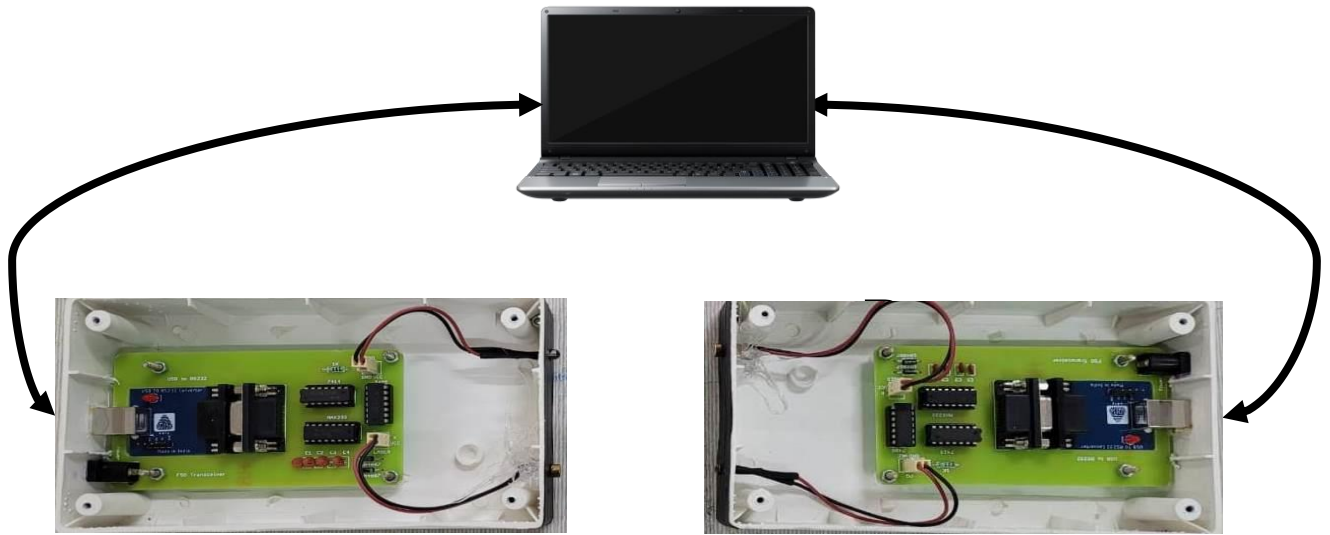
Description: An advanced platform combining ESP32 microcontroller with Raspberry Pi microprocessor capabilities for AI, ML, DL, and full-stack IoT product development.

Specifications: **Processor & Micro controller:** Raspberry Pi 4 & ESP32, **Network Interfaces:** Wi-Fi, Bluetooth, GSM, LoRa, Ethernet, FSO, **Sensors:** Pi Camera, DHT11, BMP180, IMU, Rain Sensor, Ultrasonic, **Actuators:** Relay, Speaker, Buzzer, **Display:** OLED, **Storage:** SD Card.

Number of Experiments: Number of Experiments: 63+ (Refer to list) **Documents Included:** Experiment Guide, Python Scripts, Setup Instructions **Applications:** AI-based image classification, Face recognition, Smart surveillance, IoT device control via ML

Product 6: VLC/FSO/Hybrid Optical Communication Kit





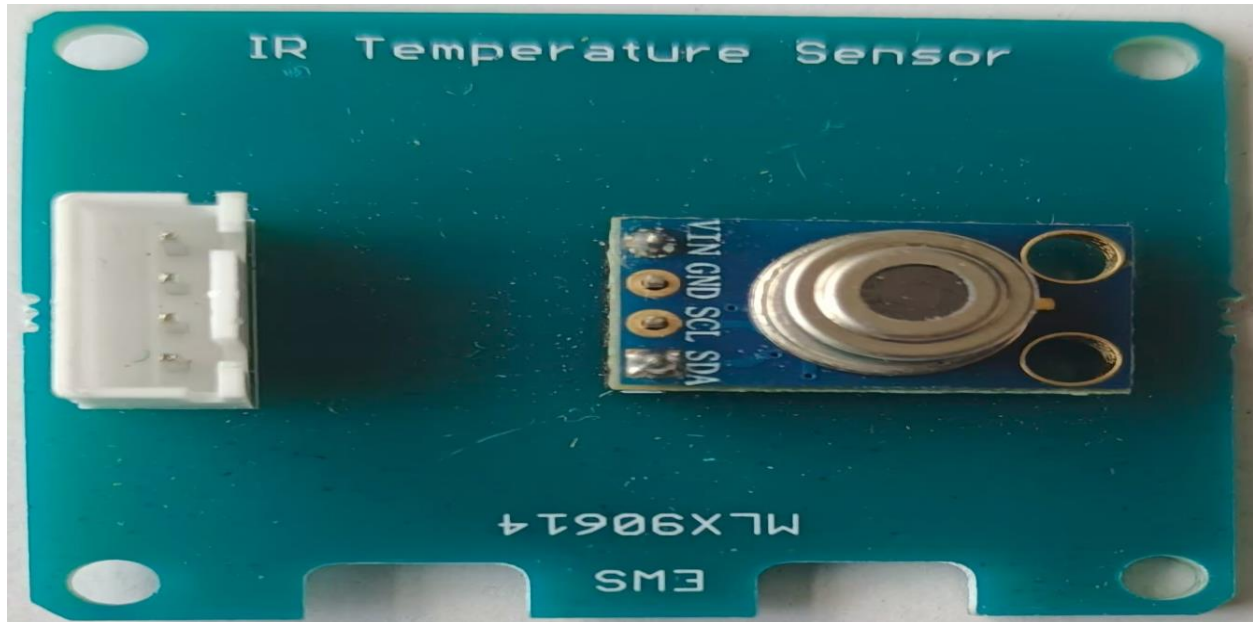
Description: An optical communication lab package designed for exploring Li-Fi, FSO, hybrid RF-optical links, and atmospheric simulation experiments.

Specifications: VLC data transmission, Image transmission over FSO, Fog simulation, Hybrid RFFSO switching

Use/Applications: These experiments provide a strong foundation in optical wireless communication by introducing the fundamentals of VLC, Li-Fi, and FSO systems using LEDs, photodiodes, and optical transceivers. Learners gain insights into how environmental factors such as wavelength, ambient noise, fog, and atmospheric turbulence influence the performance, speed, and reliability of communication links. They also develop hands-on skills in system design and prototyping, from establishing basic communication links to transmitting text and images, as well as implementing hybrid FSO/RF systems with automatic switching mechanisms. In addition, the experiments emphasize performance evaluation and analysis, enabling students to measure and interpret parameters such as transmission rate, stability, visibility, and PSNR under different controlled conditions.

Grove Sensor Modules

Grove IR Temperature Sensor

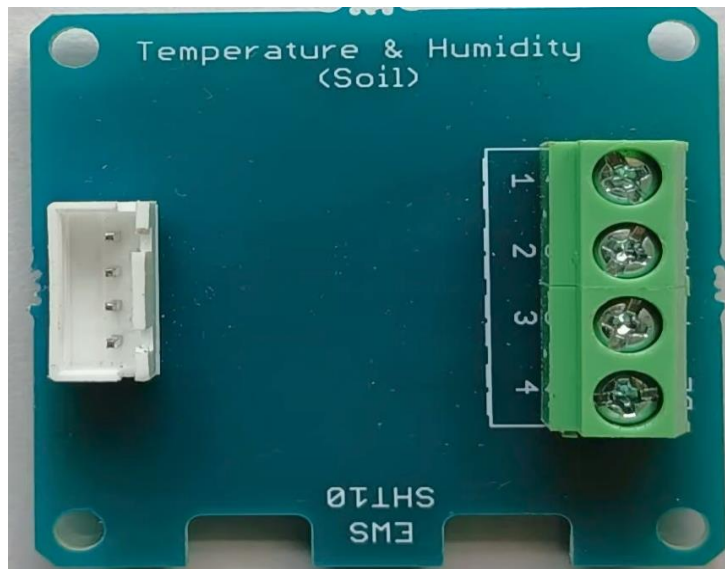


Description: This is a versatile and highly efficient non-contact infrared (IR) thermometer sensor designed for accurate temperature measurements of objects and surfaces without physical contact. It operates by detecting the infrared radiation emitted by a target, making it ideal for a wide range of applications.

Specifications: Based on its core function of precisely measuring surface temperature from a distance, this sensor features a wide object temperature range from $-70\text{ }^{\circ}\text{C}$ to $+380\text{ }^{\circ}\text{C}$ ($-94\text{ }^{\circ}\text{F}$ to $+716\text{ }^{\circ}\text{F}$), supporting targets from the very cold to the extremely hot, and it operates reliably in harsh environments with an ambient temperature range of $-40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$. It delivers high accuracy of $\pm 0.5\text{ }^{\circ}\text{C}$ within the common $-20\text{ }^{\circ}\text{C}$ to $+120\text{ }^{\circ}\text{C}$ range and offers dual output options: a digital SMBus/I²C interface (default address 0x5A) for easy microcontroller integration and an optional analog PWM output. The sensor is ideal for battery-powered devices due to its low power consumption of $\sim 1\text{ mA}$ (active) and $\sim 3\text{ }\mu\text{A}$ (sleep), and it provides a fast response time of typically under 0.5 seconds for real-time monitoring. Furthermore, it features a flexible 35° field of view (with other models available) and is compatible with 3.6 V to 5 V DC systems. Use/Applications: Industrial Monitoring & Safety, Medical and Public Health Screening, Automotive Diagnostics, Consumer Appliances & IoT, Thermal Imaging & Scanning.

Use/Applications: Industrial Monitoring & Safety, Medical and Public Health Screening, Automotive Diagnostics, Consumer Appliances & IoT, Thermal Imaging & Scanning

Grove Temperature & Humidity Module

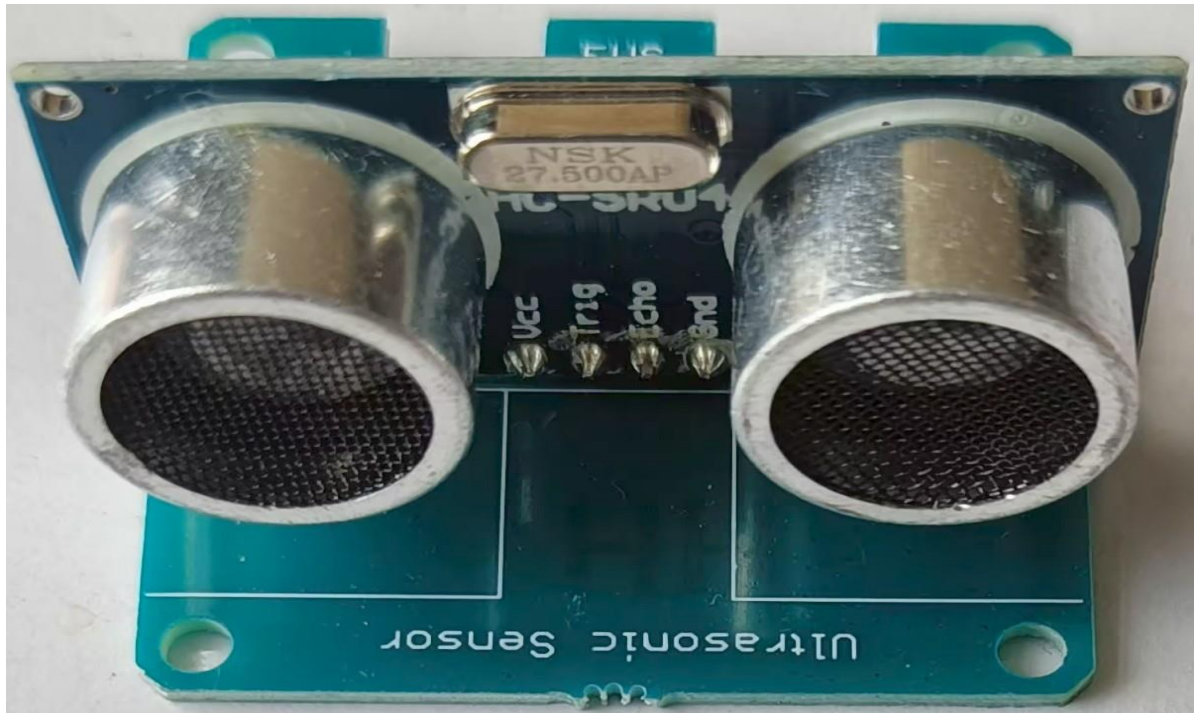


Description: The SHT-10 is a robust, digital soil sensor designed for direct, simultaneous measurement of both temperature and moisture in various media, most commonly soil. It provides reliable data for environmental monitoring and control systems with high accuracy and low power consumption.

Specifications: This sensor integrates digital measurement of temperature and relative humidity for soil or other media, featuring a broad temperature range from -40°C to $+123.8^{\circ}\text{C}$ and a full 0% to 100% Relative Humidity (RH) range, with high accuracy of $\pm 0.5^{\circ}\text{C}$ for temperature and $\pm 3.5\%$ RH for humidity. It uses a proprietary 2-wire digital interface (Clock and Data) for stable, long-distance data transmission, though it is not I²C compatible and requires a specific driver. Operating from 2.4V to 5.5V DC with a low typical current of $\sim 550\ \mu\text{A}$, it is ideal for battery or solar-powered applications, and it typically requires 4 seconds for humidity readings to stabilize, commonly available as a module for easy integration. Price Plan in Indian Rupees (Rs.) excluding GST and transport: As of the number of units (minimum order 2) (one-day online training free), offline training will be chargeable.

Use/Applications: Precision Agriculture & Smart Farming, Greenhouse & Nursery Automation, Environmental & Weather Monitoring, Landscaping & Golf Course Management, DIY Smart Gardening.

Grove Ultrasonic Sensor



Description: The HC-SR04 is a popular and cost-effective ultrasonic distance measurement module. It operates on the principle of sonar, calculating the distance to an object by emitting an ultrasonic pulse and timing how long it takes for the echo to return. This provides a reliable, non-contact method for ranging and object detection.

Specifications: This sensor operates on the principle of non-contact ultrasonic ranging using 40 kHz sound waves, with an impressive measurement range from 2 cm to 400 cm suitable for both close-proximity and medium-range applications. It provides high precision with an accuracy of ± 3 mm and a 1 mm resolution, while its focused 15-degree cone-shaped beam minimizes false readings from surrounding objects. Featuring a simple control interface, it is triggered by a 10 μ s TTL pulse and outputs a TTL pulse with a width proportional to the distance, operates on a 5V supply with low power consumption (~ 15 mA), and offers easy integration via a standard 4-pin interface (VCC, Trig, Echo, GND).

Use/Applications: Robotics & Automation, Level & Distance Monitoring, Security & Alarms, DIY Projects & IoT

Grove Buzzer Sensor

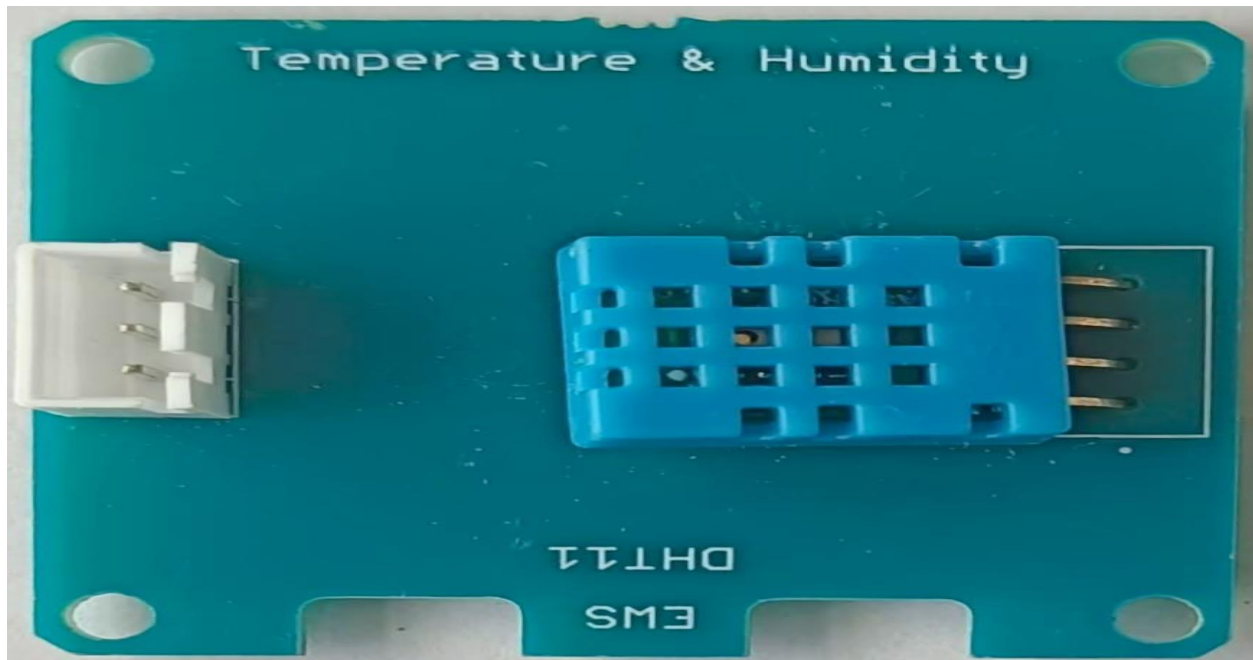


Description: This versatile buzzer module is an essential audio feedback component for electronics projects, offering two distinct operating modes in a single, easy-to-use package. Its wide operating voltage and compatibility with popular development platforms make it a go-to solution for adding sound alerts, alarms, and even melodies.

Specifications: This module offers dual-mode operation as both an active and passive buzzer: the active mode contains an internal oscillator to produce a continuous, single-tone sound at ~ 2.5 kHz when powered, ideal for simple alerts, while the passive mode requires an external drive signal to generate various tones, melodies, and sound effects via PWM. It delivers a loud, fixed-frequency output of approximately 85 dB in active mode, and a fully adjustable frequency in passive mode, enabling the playback of simple tunes. With an operating voltage of 3.3V to 5V DC and low current consumption of 10–35 mA, it is directly compatible with popular microcontrollers and comes in a convenient pre-wired module with a standard 3-pin header for easy integration.

Use/Applications: Use the ACTIVE Buzzer for simple, single-tone alerts and alarms, Use the PASSIVE Buzzer for complex tones, melodies, and multi-sound feedback.

Grove DHT11 Sensor

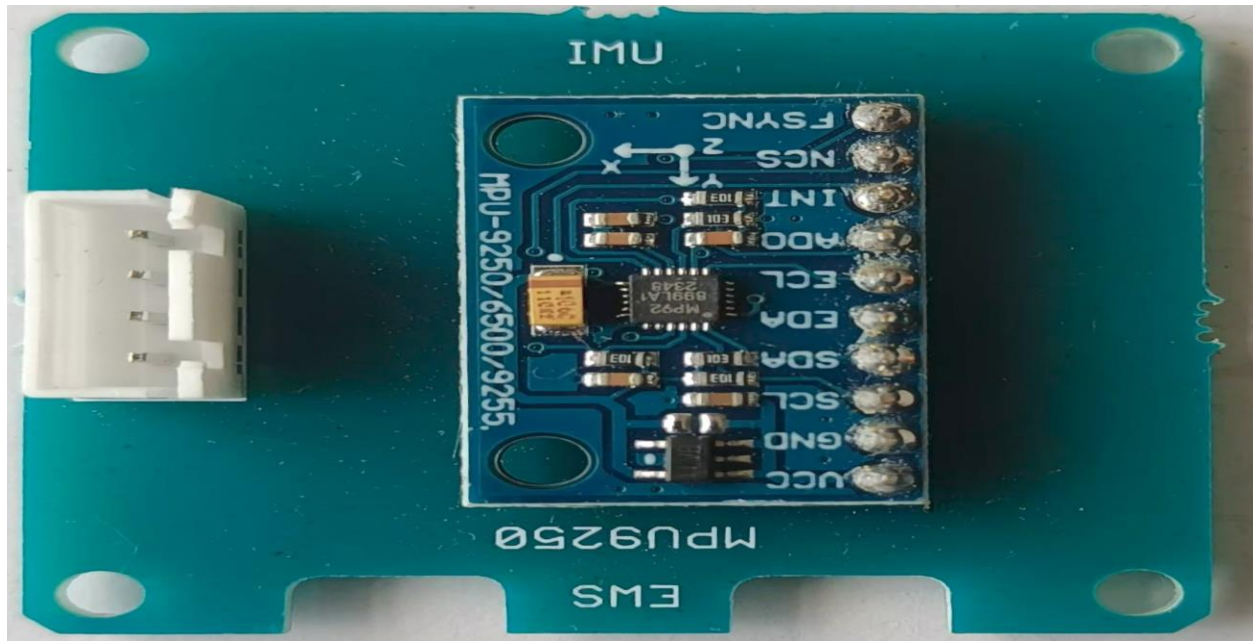


Description: The DHT11 is a low-cost, popular digital sensor that provides calibrated readings of temperature and humidity. It integrates a resistive-type humidity sensing element and an NTC thermistor, connecting to a dedicated 8-bit microcontroller for simple digital signal output. It is a go-to solution for basic environmental monitoring where high precision is not critical.

Specifications: This sensor provides combined digital readouts for ambient temperature and relative humidity, with measurement ranges of 0°C to 50°C and 20% to 90% RH, and an accuracy of $\pm 2^\circ\text{C}$ for temperature and $\pm 5\%$ for humidity. It utilizes a single-wire digital protocol for simplified microcontroller connections, operates on 3.3V to 5.5V DC with a maximum current consumption of 2.5 mA, and has a sampling rate of one reading per second, requiring less than 5 seconds for the reading to stabilize.

Use/Applications: Home Environment Monitoring, Gardening & Plant Care, Educational & DIY Electronics, Basic Weather Stations, Asset & Storage Monitoring

Grove IMU Sensor

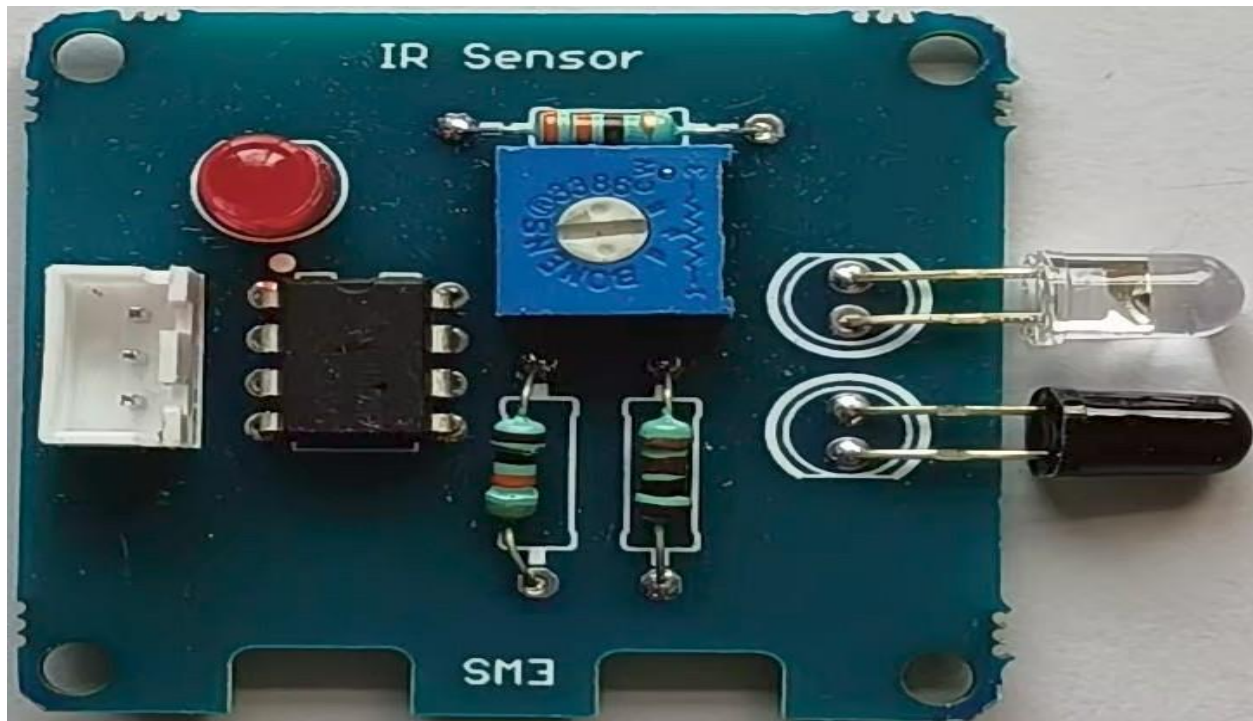


Description: This is a versatile 9-axis Inertial Measurement Unit (IMU) that integrates three critical motion-sensing technologies into a single chip: a 3-axis gyroscope, a 3-axis accelerometer, and a 3-axis magnetometer. By fusing data from all nine degrees of freedom, it enables sophisticated motion tracking, orientation detection, and heading estimation, making it a cornerstone for advanced navigation and stabilization systems.

Specifications: This 9-DOF sensor provides comprehensive motion and orientation data by measuring angular velocity with a gyroscope, linear acceleration with an accelerometer, and magnetic fields with a magnetometer. It features high-resolution sensing with programmable ranges for each sensor, supports flexible I²C and SPI communication, operates from 2.4V to 3.6V with low power consumption, and is built for robust performance across a wide temperature range from -40°C to +85°C.

Use/Applications: Robotics and Drones, Augmented and Virtual Reality (AR/VR), Navigation and Positioning Systems, Consumer Electronics, Industrial Monitoring.

Grove IR Sensor

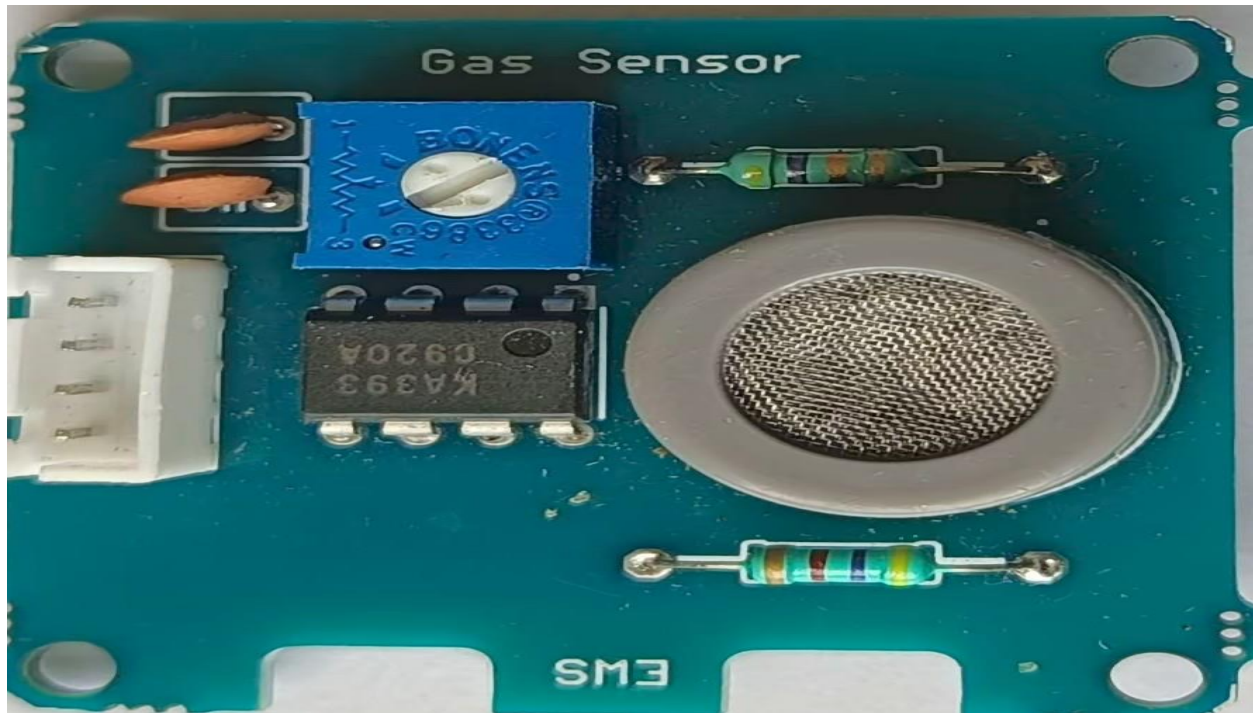


Description: This is a fundamental and widely used infrared obstacle detection sensor module. It operates on a simple principle: it emits an infrared light and detects whether that light is reflected back by a surface in front of it. The module provides a clean digital signal, making it extremely easy to interface with microcontrollers for basic presence and avoidance applications.

Specifications: This sensor detects the presence of an obstacle without physical contact using an active infrared (IR) reflection principle, where an IR LED emits light and a phototransistor detects its reflection across an adjustable range of approximately 2 cm to 30 cm. It provides a simple digital output, going LOW when an obstacle is detected and HIGH when clear, and is easy to use with an operating voltage of 3.3V to 5V DC, an on-board potentiometer for sensitivity adjustment, and a current consumption of ~20 mA for direct microcontroller compatibility.

Use/Applications: Robotics and Mobility, Industrial and Automation, Consumer Electronics and DIY Projects, Prototyping and Education

Grove Carbon Monoxide Sensor

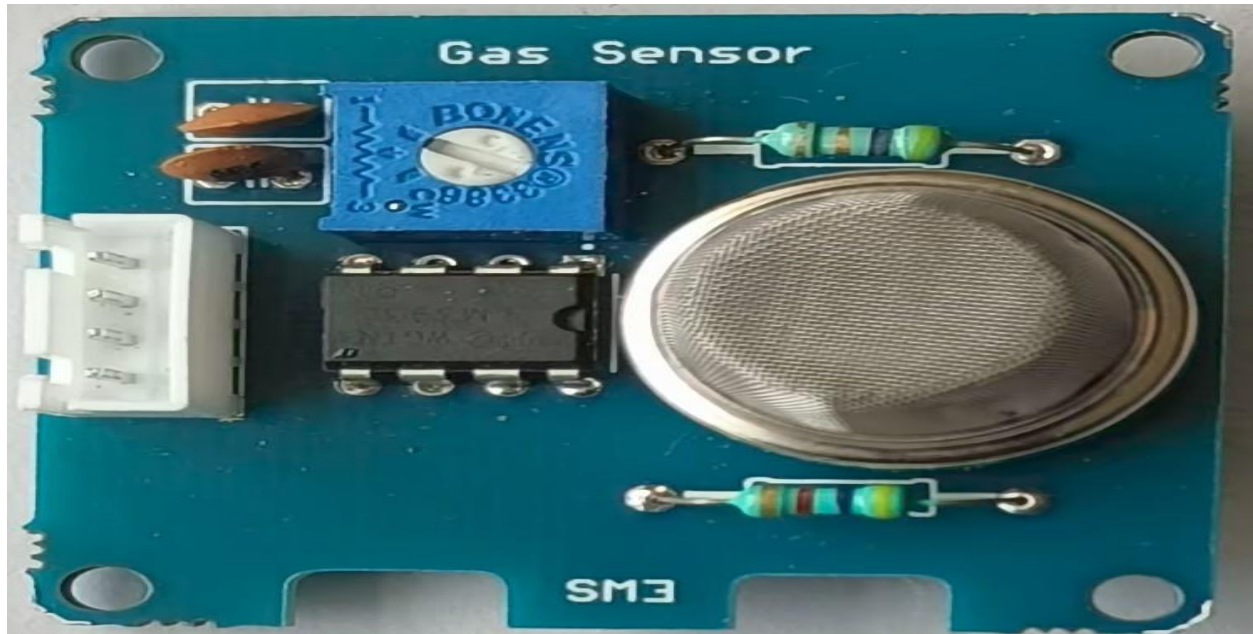


Description: The MQ-7 is a robust gas sensor module specifically designed for the detection of Carbon Monoxide (CO), a colorless, odorless, and poisonous gas. It uses a Chemi resistive sensing element made of tin dioxide (SnO_2), which has a lower conductivity in clean air. When CO gas is present, the sensor's conductivity increases, and this change is converted into an analog output voltage proportional to the gas concentration.

Specifications: This carbon monoxide (CO) sensor effectively detects concentrations from 20 to 2000 ppm with high selectivity, requiring a unique dual-heating cycle of 60 seconds at high heat and 90 seconds at low heat for optimal performance, provides an analog output for concentration estimation with a response time under one minute, but needs a 48-hour initial preheat and operates on a 5V supply with significant heater power consumption.

Use/Applications: Home and Workplace Safety, Industrial Safety and Environmental Monitoring, Portable Air Quality Monitors, Educational and Research Projects

Grove Gas Sensor

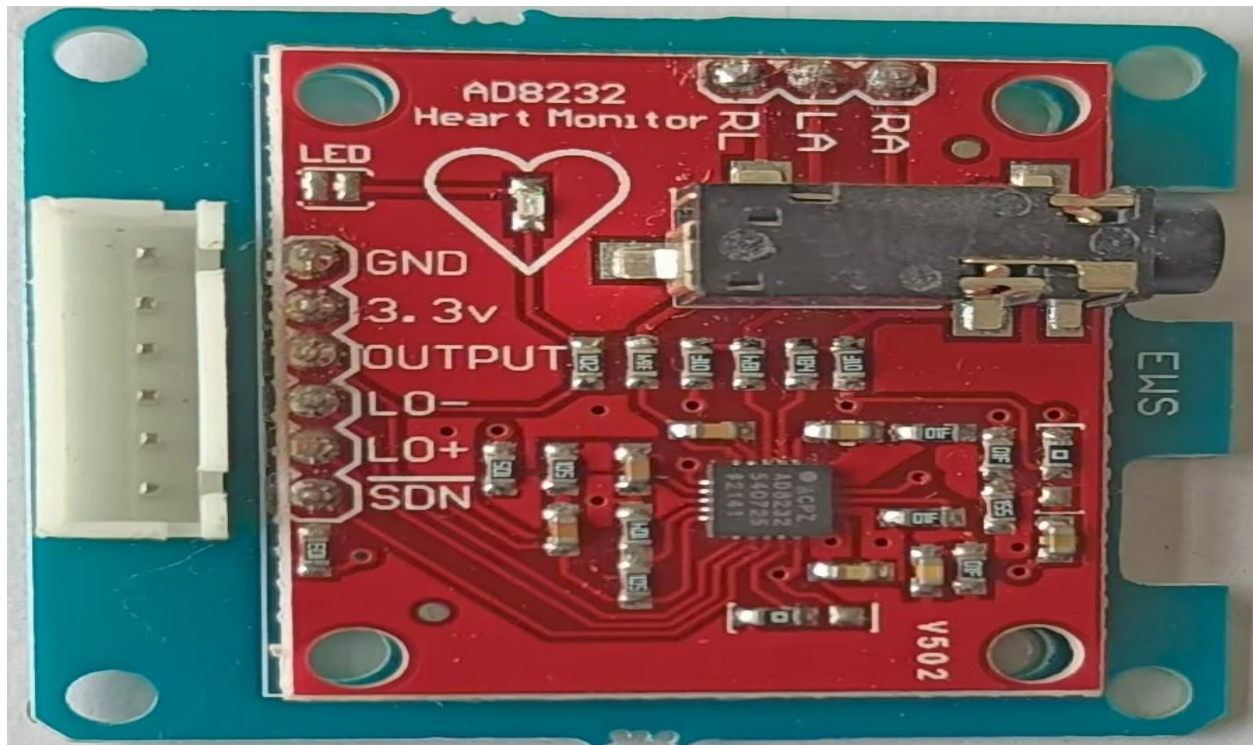


Description: The MQ-135 is a versatile and widely used gas sensor module designed for the qualitative detection of a broad range of harmful gases and pollutants in the air. It functions as a general "air quality" sensor rather than a detector for one specific gas. Its sensing element is made from tin dioxide (SnO_2), whose conductivity changes in the presence of various contaminants, producing an analog voltage signal that corresponds to the overall concentration of detected gases.

Specifications: This sensor detects a wide spectrum of pollutants including ammonia, nitrogen oxides, and benzene, providing a broad-spectrum, non-selective response to general air contamination with a detection range from 10 to 1000 ppm, delivers a fast-response analog output, but requires a 24-48 hour preheat and significant heater power (800 mW) while operating robustly across wide temperature and humidity ranges.

Use/Applications: Indoor Air Quality (IAQ) Monitoring, Ventilation Control Systems, Industrial and Agricultural Safety, DIY and Educational Projects

Grove ECG Sensor

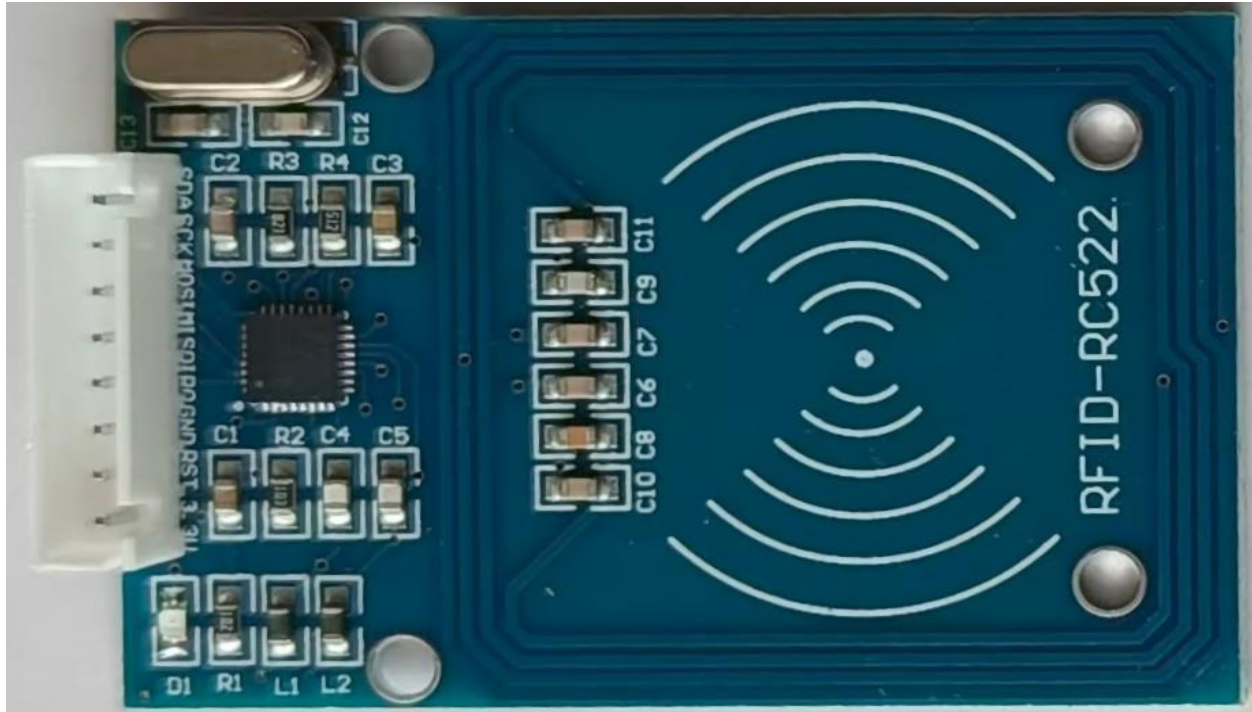


Description: The AD8232 is a specialized, integrated analog front-end module designed to reliably extract, amplify, and filter small electrocardiogram (ECG) signals in the presence of noisy conditions, such as those created by motion or remote electrode placement. It simplifies the complex task of capturing a clean heartbeat signal, making it accessible for makers, developers, and wellness applications.

Specifications: This integrated circuit acts as a complete signal conditioning block for ECG and heart rate measurement, amplifying the heart's faint electrical signals into a high-fidelity 0 to 3.3V analog output waveform that preserves key characteristics like the P-wave and QRS complex. It ensures a clean signal through advanced noise reduction, including a high CMRR to cancel interference and integrated filters, and features crucial lead-off detection, all while consuming ultra-low power of $\sim 170 \mu\text{A}$ and operating on 3.3V to 6V with a simple three-electrode setup.

Use/Applications: Personal Fitness and Wellness, Remote Patient Monitoring (RPM), Research and Prototyping, Educational Demonstrations.

Grove RFID Reader Sensor



Description: The RC522 is a highly integrated and low-cost RFID module that operates at 13.56 MHz, enabling wireless communication with compatible tags and cards. Based on the NXP MFRC522 chipset, it serves as both a reader and a writer, allowing your microcontroller projects to not only identify but also write data to RFID tags. Its low power consumption and flexible interface options make it a staple for identification and data transfer applications.

Specifications: This 13.56 MHz RFID reader/writer communicates with MIFARE and NTAG tags via inductive coupling over short distances, primarily using a high-speed SPI interface while offering configurable I²C or UART alternatives, and operates on 3.3V with low power consumption of 13–26 mA during active reading and 10 μ A in standby mode.

Use/Applications: Access Control and Security, Inventory and Asset Management, Interactive Systems and Payments, Smart Home Automation

Stepper Motor Grove Module



Description: A stepper motor is a brushless, synchronous electric motor that converts digital pulses into precise mechanical shaft rotation. Unlike standard DC motors that spin freely, stepper motors move in discrete "steps," allowing for exceptional control over position and speed without the need for a closed-loop feedback system (like an encoder), though encoders can be added for verification.

Specifications: Operating on the principle of dividing a full rotation into discrete steps, this stepper motor provides precise angular positioning with common step angles of 1.8° or 0.9° , delivers high holding torque and strong low-speed performance, requires a dedicated driver for operation, and supports micro stepping for smoother, higher-resolution motion.

Use/Applications: Precision Positioning and Automation, Office and Consumer Electronics, Medical and Laboratory Equipment, Textile and Packaging Machinery.

Potential of Hydrogen Sensor Grove Module



Description: This module is an interface for a pH electrode, designed to measure the hydrogen-ion activity in a water-based solution, determining its acidity or alkalinity. The Grove system simplifies the connection, making it easy to integrate precise pH measurement into microcontroller projects without complex analog circuit design. It translates the minute analog voltage from a standard pH probe into a readable signal.

Specifications: This sensor measures the pH level of a liquid from 0 to 14 via an analog output, is compatible with standard systems through its Grove and BNC connectors, and features an onboard potentiometer for calibration while operating on 3.3V or 5V DC.

Use/Applications: Hydroponics and Precision Agriculture, Environmental and Water Quality Monitoring, Educational and DIY Science Projects, Industrial Process Control.

Turbidity Sensor Grove Module



Description: This module is designed to measure the turbidity of a liquid, which indicates the cloudiness or haziness caused by suspended solids. It operates on the optical principle, where a light source shines through the water, and a photodetector measures the amount of light scattered by the suspended particles. The Grove system provides a simple plug-and-play interface, making it easy to integrate water quality monitoring into electronic projects.

Specifications: This sensor measures the relative turbidity of water from 0 to 1000 NTU using an infrared LED and a phototransistor to detect scattered light, provides an analog voltage that decreases with higher turbidity, and connects easily via a Grove plug for operation on 3.3V or 5V DC.

Use/Applications: Water Quality and Environmental Monitoring, Industrial and Process Control, DIY Smart Home and Gardening, Educational and Citizen Science.

Total Dissolved Solids sensor Grove Module



Description: A Total Dissolved Solids (TDS) sensor is an electronic instrument that measures the combined content of all inorganic and organic substances contained in a liquid that are molecularly dissolved. It operates on the principle of electrical conductivity (EC). Since dissolved ions increase water's ability to conduct electricity, the sensor measures the conductivity and approximates the TDS value (in parts per million, ppm) using a conversion factor.

Specifications: This sensor measures the concentration of dissolved solids in water from 0 to 1000 ppm by measuring electrical conductivity and applying a conversion factor, provides both analog and digital outputs, features automatic temperature compensation for accuracy, uses corrosion-resistant electrodes, and is compatible with common platforms via a 3.3V or 5V DC supply.

Use/Applications: Water Quality and Purification, Hydroponics and Agriculture, Food and Beverage Industry, Swimming Pool and Spa Maintenance.

Rain Sensor Grove Module

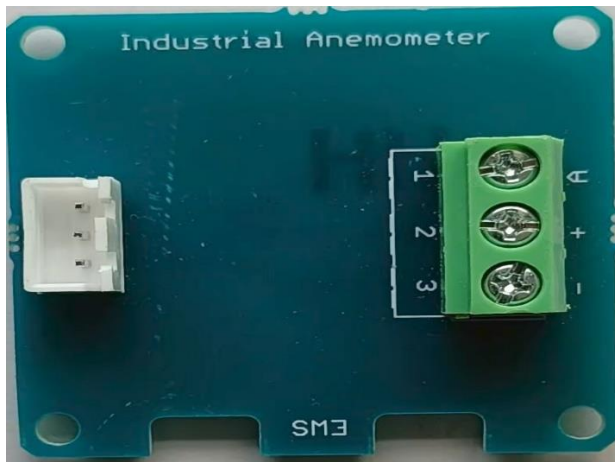


Description: A rain sensor is a simple, yet effective, module designed to detect the presence and intensity of rainfall or moisture. The typical module consists of a sensing board that is exposed to the environment and a separate control board that processes the signal. It operates on the principle of electrical resistance: when water bridges the conductive traces on the sensing board, it completes a circuit, significantly lowering the electrical resistance between the traces.

Specifications: This sensor detects water presence and measures rainfall intensity by measuring the decrease in electrical resistance across its traces, providing both an analog output for intensity and a digitally triggered output with adjustable sensitivity, and operates on 3.3V to 5V for easy microcontroller integration.

Use/Applications: Smart Home and Garden Automation, Weather Stations and Environmental Monitoring, Automotive and Transportation, Industrial and Agricultural Monitoring.

Anemometer Sensor Grove Module



Description: An anemometer is a meteorological instrument designed to measure the speed and sometimes the direction of wind. For electronic and DIY applications, these are typically compact, digital sensors that translate physical wind movement into a measurable electrical signal, allowing microcontrollers to quantify wind speed for logging and automated control.

Specifications: This sensor measures wind speed using cup, ultrasonic, or vane mechanisms, providing output through pulse, analog, or serial signals, and features a wide measurement range with durable construction for continuous outdoor use.

Use/Applications: Weather Stations and Environmental Monitoring, Renewable Energy Systems, Aviation, Drones, and Maritime, Agriculture and Construction, Smart Home and Industrial Automation

Brushless Direct Current Motor Grove Module



Description: A Brushless DC Motor is an advanced electric motor that operates on direct current but eliminates the physical brushes and commutator found in traditional DC motors. Instead, it uses an electronic controller to sequentially energize the motor's windings, creating a rotating magnetic field that turns the permanent magnet rotor. This design offers superior performance, reliability, and efficiency.

Specifications: Operating on electronic commutation for high efficiency and power density, this brushless DC motor delivers high-speed, precise control with a long, maintenance-free lifespan, requires an electronic speed controller (ESC) for operation, and its performance is defined by its KV rating which determines its rotational speed per volt.

Use/Applications: Drones and Electric Aviation, Automotive and E-Mobility, Industrial Automation and Robotics, Consumer Electronics and Appliances, Medical Devices.

Servo Motor Grove Module

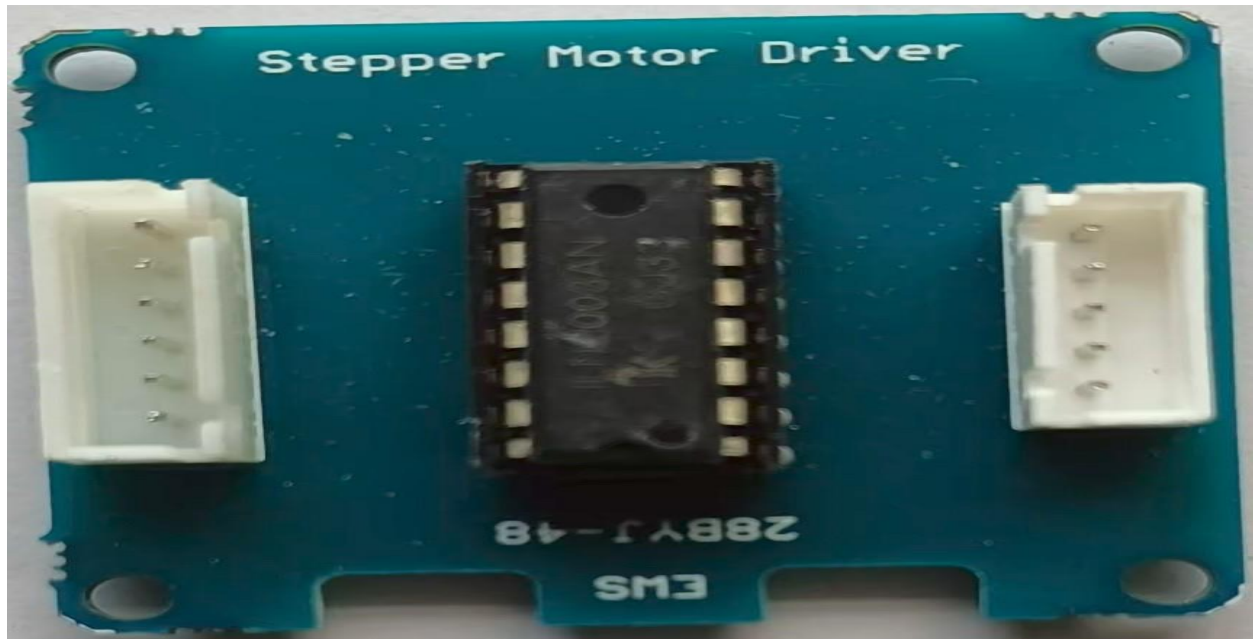


Description: A servo motor is a compact, closed-loop actuator that provides precise control over angular position, velocity, and acceleration. Unlike a standard DC motor that spins continuously, a servo is designed to move to and hold a specific position based on a coded signal. It integrates a DC motor, a gear train, a position sensor (typically a potentiometer), and control circuitry into a single, self-contained unit.

Specifications: This servo motor uses a PWM signal for precise positional control within a 0° to 180° range, delivers high torque through an integrated gearbox and closed-loop feedback system, operates via a simple three-wire interface, and is commonly available in standard, continuous rotation, and digital types with specifications like 4.8V-6V operating voltage and a torque rating of 2.5-25 kg-cm.

Use/Applications: Robotics and Automation, Radio-Controlled (RC) Hobbies, Camera and Sensor Systems, Model Making and Animatronics, Industrial and Prototyping.

Brushless BYJ Stepper Motor Grove Module

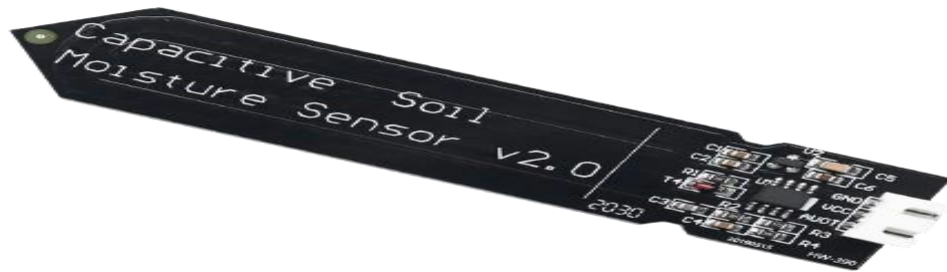


Description: The 28-BYJ48 is a small, low-cost, and widely available unipolar stepper motor. The term "Brushless BYJ Stepper Motor" is accurate, as all stepper motors are inherently brushless. This specific model is characterized by an internal planetary gearbox that provides high torque at the output shaft at the expense of speed, making it ideal for applications requiring slow, powerful, and precise movements without the need for high speed.

Specifications: This 5-wire unipolar stepper motor incorporates a 1:64 reduction gearbox for high torque, achieving 4096 steps per revolution in full-step mode and 8192 in half-step mode, operates on 5V DC, and is typically driven by a ULN2003 Darlington array module.

Use/Applications: Precision Positioning in DIY Projects, Robotics and Animatronics, Consumer Appliances and Gadgets, Fluid Control.

Soil Sensor Grove Module

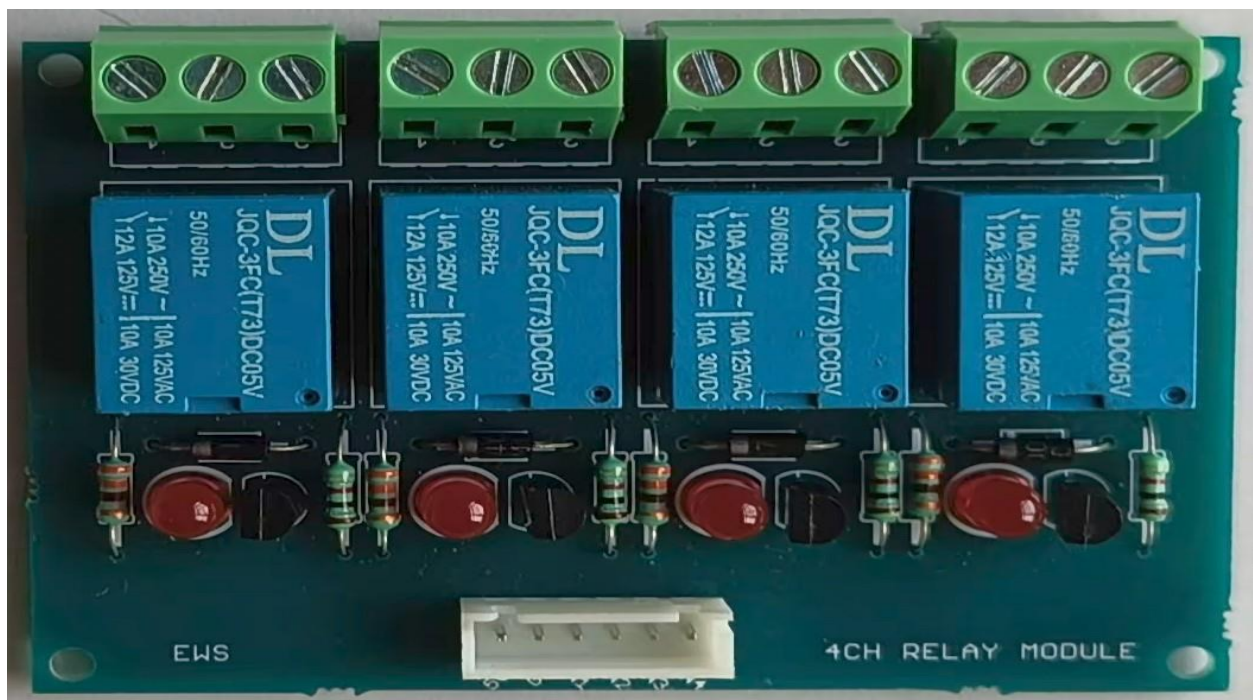


Description: A soil moisture sensor is an electronic device that measures the water content in soil. The most common types for DIY and gardening use are capacitive and resistive sensors, which estimate the volumetric water content by measuring the soil's dielectric constant or electrical conductivity, respectively.

Specifications: This sensor measures relative soil water content using either a resistive or a more accurate capacitive principle, provides both analog and digital outputs for reading moisture levels or triggering alerts, and operates on 3.3V or 5V for easy integration with platforms like Arduino and Raspberry Pi.

Use/Applications: Precision Agriculture & Smart Gardening, Landscaping and Turf Management, Environmental and Scientific Research, Consumer IoT and DIY Projects.

Grove Actuator Modules

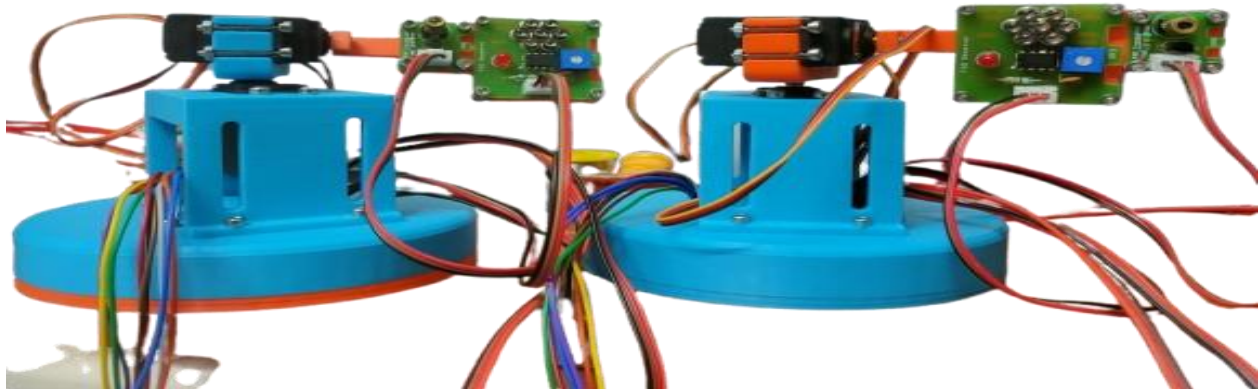


Description: A Grove Actuator is not a single device but a category of components within the Grove ecosystem that convert an electrical signal from a microcontroller into physical motion. The key feature is the standardized **4-pin Grove connector**, which makes wiring simple and foolproof by integrating power, ground, and control signals into a single plug. These actuators are the "muscles" of a project, allowing it to interact with the physical world.

Specifications: This Grove actuator module performs a physical action like movement or lighting based on a control signal, utilizes a standardized 4-pin connector for plug-and-play integration with platforms like Arduino and Raspberry Pi, and encompasses various devices including servos, relays, and vibration motors, often requiring an external power supply for the actuation itself.

Use/Applications: Robotics and Motion Control, Smart Home and Automation, Interactive Displays and IoT, Industrial Prototyping

Auto Aligned FSO / Laser Trans-Receiver Grove Module



Description: The Auto Aligned FSO / Laser Trans-Receiver Grove Module is a specialized module designed to detect the presence or absence of an object by using a beam of light transmitted through a fiber optic cable. It operates on a "through-beam" principle, where the module has separate components for transmitting and receiving light. When an object breaks the light beam traveling through the fiber optic cable, the sensor's output state changes.

Specifications: This Grove fiber optic sensor detects objects by sensing the interruption of a light beam transmitted through fiber optic cables, provides a digital output signal when triggered, and features a spring-loaded holder for easy alignment and connection via a standard 4-pin plug to operate on 3.3V or 5V DC.

Use/Applications: Industrial Automation and Counting, Hazardous and Electrically Noisy Environments, Specialized Mechanical Sensing, Security and Safety

Get in Touch

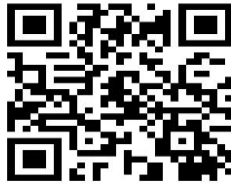
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Arnapurna Vihar, New Petrol pump lane, Near Sai Mandir, Tankapani Road, Near the calibration spare 1st Gate, Bhubaneswar, Pin-751002

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